7 Conclusion

Under the baseline rightward traction, all three clamped objects extend. The square's mean right-edge horizontal displacement is about 0.494 mm, with decreasing changes under refinement. In the imported notebook results, the rectangle of twice the length extends about twice as far, the central opening increases the mean extension by 13%, and clamping only half of the left edge increases it by 11 to 25%. Moving an equal-length load patch changes the in-plane vertical deflection. Mean extension changes little for the square and rectangle, but noticeably for the perforated square.

For these supported symmetric positive-definite systems, conjugate gradients is a practical starting choice among the tested iterative methods. CG, diagonal PCG, MINRES and full GMRES reach the required residual on all 27 distinct notebook systems. PCG reduces iterations, but is slower than CG in 26 recorded comparisons and only about 0.5% faster in the remaining one. Ordinary Jacobi diverges and damped Jacobi exhausts the budget throughout; Gauss–Seidel and steepest descent each succeed only on the coarsest square and perforated-square systems. These results favour checking residual acceptance and total cost rather than selecting a method by simplicity or iteration count alone. The untimed sparse-LU solution remains a same-mesh reference, not a measured performance competitor.

The validation checks answer different questions. Prescribed displacements, force and moment balance, and agreement with the direct reference support the discrete solution. Refinement tests mesh dependence; the separate exact affine test checks a different, roller-supported case. Neither establishes the continuum error of the clamped objects. Conditioning worsens under refinement even while the mean displacements settle. The structured-square perturbation tests also show that sensitivity depends on how the load changes: the condition number describes possible amplification, not the response to every perturbation.

These conclusions are limited to the small-deformation, plane-stress model, the tested cases and the recorded timing protocol. The opening's larger remaining mesh dependence motivates further refinement, not a claim that corner stresses caused it. Useful next steps are to test stronger preconditioners with setup included, examine local refinement and stress behaviour, and check the physical model's validity. No material or thickness comparison has been made.